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Investments in renewable energies by German households: A matter of economics, social influences and ecological concern?



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ABSTRACT

We analyzed how internal factors (e.g., value orientations or attitudes) and external factors (e.g., incentives or social influences) affect households' decisions to make an investment in solar thermal energy, and how these factors interact in the decision process. 232 German home owners were questioned using an online survey. Internal factors were measured in the form of value orientations. The influence of external factors was measured in a discrete choice experiment, which covered economic consequences, social influences and ecologic consequences. The data was analyzed using mixed logit models. We found decision makers with strong eco-social value orientations to be more likely to make an investment while strong conservative and strong hedonistic value orientations both reduced investment likelihood. Among the external factors, economic consequences and investment recommendations by trustworthy sources were found highly important. Ultimately, we found several interactions between value orientations and external factors: decision makers with strong conservative or strong hedonistic value orientations are more sensitive towards economic consequences; decision makers with strong eco-social value orientations, however, are more sensitive towards recommendations and ecological consequences of an investment. Our results propose that strategies fostering households' investments in renewable energies should be appropriately adjusted for different groups of decision makers.

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1. Introduction

1.1. The share of renewable energies in heat production is still low

Current energy production and consumption have led to severe global problems (e.g., climate change, permanent disposal of nuclear waste). Coping with these problems is one of the most pressing tasks society faces today and in the future [1]. To restrict the global temperature increase to the agreed limit of two degrees Celsius, worldwide decarbonization has to be accomplished by 2070 at the latest.

Some positive developments have taken place already: during the last decades, the use of renewable energies has increased substantially. This increase, however, mainly concerns electricity rather than heat generation. In 2014, about 20% of worldwide electricity generation was based on renewable energy sources. In contrast, the share of renewable heat generation was not even

4% [2]. Consequently, the worldwide energy generation is still mainly based on gas (47%), (non-renewable) solid fuels (39%) and petroleum (5%; [2]). Thus, increasing the share of renewable energies in heat generation should be one main goal in energy policies.

Especially in Western Societies, a large amount of final energy is used in households [2,3]. In this sector, most energy is used for space heating (53%) and water heating (16%)—what results in substantial CO₂ emissions [3]. These emissions could be reduced considerably if households decided to install systems producing heat from renewable energy sources (e.g., Ref. [4]). Solar thermal energy systems are one promising technology households can purchase for that matter. These systems can be used for both, water heating and space heating—in case the house's heating system is based on water circulation which is most common in Western Countries (see e.g., Refs. [5,6] for more information on heating systems in Germany). The residential use of solar thermal energy systems holds substantial CO₂ (and energy cost) saving potentials—especially if they are used for both water and space heating (e.g., Refs. [4,7]). Calculations for German households show that, for instance, in single- and two-family houses more than 25% of the heat energy demand could be covered by solar thermal power

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[4]. Currently, solar thermal energy systems are installed in only about 10% of these houses, though [8]. If the use of solar thermal energies was expanded to all single- and two family houses in Germany, an additional 108 TWh of heat energy and 25 Gt of CO₂ emissions could be saved per year (energy saving potentials were calculated from average consumption data [5,6,9]).

The present study aims at investigating how household decision makers get motivated to make – or not to make – investments in solar thermal energy systems and how different factors interact within the decision process. Our main goal is to provide insights that could help fostering such investments. We focus on owner-occupied residential buildings where investors are directly affected by their decisions [10]. Our study was conducted in Germany but the findings are likely applicable to other Western countries where circumstances are similar and comparable measures are taken to further household investments in renewable energies. In fact, the findings may be even more relevant for other countries: for one thing, the share of renewable sources used for German heat production is approximately 10% [11]—which is a rather high percentage compared to other countries (see above [2]). For another thing, even though owner occupation is high in Germany (53.3%; [12]) it is still low as compared to other Western Countries (e.g., 70.1% in the European Union on average or 64.5% in the United States [12,13]).

1.2. How home owners decide to make investments in renewable energies

As any other behavior, making investments in solar thermal energy systems is based on various factors. According to the ABC theory by Guagnano et al. [14] behavior (B) is a function of internal factors (A) and external factors (C). *Internal factors* involve any kind of *dispositions* (e.g., attitudes, value orientations or personal norms) motivating an individual to act in a certain way. For instance, these dispositions are egoistic (e.g., as one seeks for wealth or influence), eco-social (e.g., as one seeks to protect the natural environment and society in general) or hedonistic (e.g., as one seeks for immediate satisfaction; see e.g., Refs. [15–18] for more detailed information on dispositions). Sometimes different dispositions suggest the same behavior [19]. A decision maker may, for example, make an investment in solar thermal energy systems due to both eco-social (e.g., to protect the natural environment) and egoistic dispositions (e.g., to save money). Other times, different dispositions can be in conflict with each other suggesting different behaviors: One may consider making an investment due to eco-social dispositions, while hedonistic dispositions may suggest spending the money otherwise (e.g., for immediate satisfaction).

External factors are related to the behavior context determining *behavior difficulty*. Among others, these factors cover the degree of financial, social or structural support decision makers' experience. A certain behavior is perceived as difficult (or "costly") if it, for instance, requires a lot of money and effort or if it is rejected by the social environment. Behavior difficulty can be alleviated if financial incentives or social support are provided. The easier a behavior is perceived to be the more likely it is to be shown.

Empirical findings by Guagnano et al. [14] and other research groups suggest that internal and external factors interact (see Ref. [20] for an overview). A common finding is that perceived behavioral difficulty moderates the influence a person's dispositions have on behavior. Diekmann and Preisendörfer [21] investigated such moderation effects by analyzing the influence of eco-social dispositions on several environmentally relevant behaviors at varying difficulties. In their study, they defined seldom performed behavior as difficult. For instance, recycling behavior was found to be carried out frequently and thus classified as easy, while curtailing car use was a comparatively rare behavior which was classified

as difficult. Diekmann and Preisendörfer's findings clearly suggest that internal factors have the greatest influence on behavior if said behavior is perceived to be easy; for more difficult behaviors their influence decreased (*low cost hypothesis*). However, it should be pointed out again that – in line with several other studies – Diekmann and Preisendörfer only accounted for internal factors in terms of eco-social dispositions while other potentially relevant dispositions (e.g., egoistic or hedonistic ones) were not included.

Interaction effects between internal and external factors were also investigated in a similar study by Guagnano et al. [14]. Here it was analyzed how the influence of eco-social dispositions on (environmentally relevant) behavior was moderated by behavioral difficulty. Here, the authors found a reversed U shaped moderation. Thus, their findings suggest that internal factors have the greatest influence on behavior if behavioral difficulty is in the middle range: The influence of internal factors was found to be lower if behavioral difficulty was relatively high – which is in line with the low cost hypothesis – but also if behavioral difficulty was relatively low (see also Ref. [19]).

Installing a solar thermal energy system to a residential building goes hand in hand with several consequences: among others, the installation costs several thousand euros, brings about changes to the heating system as well as to the house's appearance (e.g., Refs. [4,22]). Thus, making investments in solar thermal energy systems can be classified as difficult behaviors. According to the low cost hypothesis and the ABC theory, external factors should be more relevant for these behaviors than internal ones [14,21]. Empirical research on household investments in solar power and heating systems supports this assumption: even though a large number of internal factors was considered across these studies, only a fraction of them has shown to be decision-relevant ([23–27]; see Table 1 for an overview of decision-relevant internal factors). External factors, on the other hand, were found to be more relevant in the decision process [26–33]. In most empirical studies external factors were measured in terms of perceived investment circumstances and consequences. As theory suggests, making investments was found to be more likely if decision makers perceived behavior difficulty to be reduced by certain external factors. Highly relevant external factors involved expected financial costs and benefits, expected reliability and operation security of the new system, perceived social support and recommendations by others to make an investment, and expected environmental benefits arising from the investment (see Table 1 for an overview).

It should be noted, though, that the total amount of empirical research on energy-relevant investment decisions in households is fairly limited up to now [34]. The available investigations frequently suffer from some shortcomings: for one thing, most of the available studies focus either on internal or on external factors while they do not take the interaction of these factors into account [34]. For another thing, studies accounting for internal factors mostly focus on eco-social dispositions while other potentially relevant (e.g., egoistic or hedonistic) dispositions are hardly considered (e.g., Refs. [19,35]). Thus, there are still uncertainties on how any other of these other dispositions influence energy-relevant investment decisions and how they interact with external factors [34].

Analyzing such interactions could be most useful for identifying different target groups of households—and for tailoring policy measures accordingly [19,36]. It can be argued that home owners with, for instance, strong egoistic dispositions are particularly sensitive towards financial consequences (and thus financial incentives) when it comes to solar thermal energy investments, while others with high eco-social dispositions could be more interested in environment-related investment consequences (and, for instance, environmental certifications).

Table 1

Internal and external factors that had empirically been found to influence households' investments in solar power and heating systems.

Factors	Studies
Internal factors	
Eco-social value orientation	Kastner et al. [25]
Egoistic value orientation (–) ^a	Kastner et al. [25]
Attitudes towards renewable energies	Welsch and Kühling [26]
Attitudes towards nuclear power (–) ^a	Welsch and Kühling [26], Wortmann [23]
External factors	
Expected financial consequences	Achtnicht and Madlener [28], Hübner and Müller [29], Labay and Kinnear [27], Mahapatra and Gustavsson [31], Michelsen and Madlener [32], Sadler [30], Wortmann [23]
Expected contribution to environment protection	Achtnicht and Madlener [28], Hübner and Müller [29], Mahapatra and Gustavsson [31]
Expected reliability/operation security	Achtnicht and Madlener [28], Labay and Kinnear [27], Mahapatra and Gustavsson [31], Michelsen and Madlener [32]
Perceived social support and investment recommendations by others	Achtnicht and Madlener [28], Labay and Kinnear [27], Murphy [33], Welsch and Kühling [26]

^a Negative influence: stronger attitudes or value orientations reduce investment likelihood.

1.3. Research agenda

In the present study we analyze how internal and external factors interact in influencing households' investment decisions in solar thermal energy systems. In contrast to prior studies, different kinds of internal factors are considered. In addition to eco-social dispositions, other dispositions are taken into account, such as egoistic and hedonistic orientations of the decision makers. As for external factors, we focus on decision consequences that have been proven decision-relevant (see Table 1), and that can be targeted by policy measures in general (e.g., financial aspects, guarantees) in order to ensure practical applicability.

The research design is explorative since it cannot clearly be hypothesized in advance how the analyzed factors interact. However, some interactions are expected to be more likely than others. For instance, the influence of internal factors such as egoistic dispositions might depend more strongly on external factors like expected financial consequences than on expected environment-related consequences. An opposite relationships could exist for internal factors like eco-social dispositions.

2. Method

In early 2014, we conducted an online survey with German home owners living in owner-occupied single- or two-family houses. The questionnaire was provided in a web portal operated by a market research institute. Only home owners living in their house for five years or more were invited to participate. We chose this sample because these home owners had very likely already considered making investments in renewable energy systems. Altogether, we gathered 232 data sets fitting our criteria.

The investigation was divided into two sections: at first, a discrete choice experiment was conducted to measure the influence of external factors in making investments in solar thermal energy systems. The second part included the measurement of internal factors and a number of supplementary questions. Both parts of the study will be presented subsequently.

2.1. The discrete choice experiment

We conducted a discrete choice experiment to investigate the decision-relevance of several external factors. This approach was chosen as it allows simulating a fairly realistic decision situation where some expected decision consequences – that are affected by external factors – need to be chosen over others (e.g., Refs. [37–39]).

Participants were asked to imagine the situation that their house needed a new heating system, and that the new system could partly be powered by solar thermal energy. The participants were asked

to repeatedly decide between four options: one of three different solar thermal systems and a fourth non-investment option. The three solar options differed in five *attributes* representing external factors. The survey was limited to 17 repeated choices (*choice sets*) per participant in order to avoid fatigue effects (see e.g., Ref. [40]). A choice set example is provided in Fig. 1.

Five attributes representing external factors were involved. We confined the decision problem to that number as humans generally base their decisions on a limited number of factors whereas others are widely neglected (*bounded rationality*; see Refs. [41,42]). Our attribute selection was based on two considerations: firstly, we focused on external factors that have been found decision-relevant beforehand (see Table 1). In addition, we focused on attributes that can be targeted by policy measures (e.g., via financial incentives or in communication strategies). Each attribute was scaled at different levels based on real-life values (see Table 2 for an overview). In the following, these attributes will be presented in detail.

Installation costs and *energy cost saving potentials* were included given the decision-relevance of expected financial consequences. Both attributes were scaled at three levels: Installation costs at € 5000, € 7500 or € 10,000; energy cost saving potentials at 10%, 20% or 30% p.a. as compared to the household's current energy consumption. The attribute levels were based on data provided by the German Ministry for Transport, Construction and Urban Development [9], the German Energy Agency [4] and German building associations [22].

CO₂ saving potentials were included and also scaled at 10%, 20% or 30% p.a. given the decision-relevance of expected contribution to environment protection. Naturally, CO₂ savings depend on reducing the use of fossil fuel. Therefore, CO₂ saving potentials and energy costs are correlated under realistic circumstances. Nevertheless, we treated both saving potentials as if they were unrelated in order to investigate the decision-relevance of expected financial and environment-related consequences independently of each other.

In addition, the *guarantee extent* was varied across the investment alternatives. We chose this attribute as the solar thermal systems' expected reliability was found decision-relevant before (see Table 1). Expected reliability might increase as comprehensive guarantees are provided. Guarantee extent was scaled at four levels: removal of defects for 2, 5 or 10 years, and a combined guarantee including the removal of defects and a minimum system performance (that is a minimum amount of heat production) for 10 years. These levels were chosen to match the two year guarantee period specified as the minimum by German law. Some manufacturers might offer longer periods due to their own option. Offering a performance guarantee is not usual for solar thermal systems but common for photovoltaic systems [43]. We integrated this attribute

If there are only these options, which one would you choose?

Recommendation of	Internet forum (specialized in solar thermal systems)	Journal specialized in solar thermal energy	Friends who already purchased a solar thermal system	
Investment costs	€ 5.000	€ 10.000	€ 7.500	
Energy Cost Saving Potential p.a.	30%	20%	10%	
CO ₂ Saving Potential p.a.	10%	30%	20%	
Guarantee Extent	5 years removal of defects	2 years removal of defects	10 years removal of defects	
	☐	☐	☐	☐
				I am not convinced by any of the offers. So I will not install a solar thermal system.

Fig. 1. Example of a choice set in the discrete choice experiment. **Note:** the choice set was translated for this paper. The original experiment was conducted in German.

Table 2

Attributes and levels in the discrete choice experiment.

Attributes	Levels
Investment Costs	€ 5000 € 7500 € 10,000
Energy Cost Saving Potential p.a.	10% 20% 30%
CO ₂ Saving Potential p.a.	10% 20% 30%
Guarantee Extent	2 years removal of defects; 5 years removal of defects; 10 years removal of defects; 10 years removal of defects + guaranteed minimum performance
Recommendation trustworthiness ^a	The craft business doing the houses' heating system maintenance; Nationally certified energy consultant; Friends who already purchased a solar thermal system; Journal specialized in solar thermal energy; Internet forum (specialized in solar thermal systems)

^a Investment alternatives were recommended by one of the five information sources. Participants rated the perceived trustworthiness of each information source subsequent to the decision experiment on a 5 point response scale (1=I do not trust the information source at all; 5=I completely trust the information source). Mean recommendation trustworthiness was calculated for each information source. Information sources are listed by perceived trustworthiness in descending order: the craft business was rated to be most trustworthy followed by energy consultants, friends, specialized journals and internet forums.

to investigate whether such a supplementary offer could also be promising for solar thermal energy investments.

Finally, we integrated an attribute to cover perceived social influences in the decision process. Prior studies on energy-relevant investments indicate that social influences may be exerted through *recommendations* by others [44,45]. The decision-relevance of these recommendations, however, depends on whether the information sources are perceived to be trustworthy; if an information source is perceived to be trustworthy the decision to make an investment may be triggered. In the decision experiment we specified five different information sources recommending the investment alternatives: Friends who already purchased a comparable solar thermal system, a (certified) energy consultant, the craft business currently doing the houses' heating system maintenance service, a journal specialized in solar thermal energy, and an internet

forum for experienced users of solar thermal systems. The perceived *recommendation trustworthiness* of each information source was measured in the second part of our investigation. Participants were asked to rate how much they trusted each source giving reliable investment recommendations using a 5 point response scale (1=I do not trust the information source at all; 5=I completely trust the information source; based on [46]). Across all participants, the craft business was found to be the most trusted information source ($M = 4.01$; $SD = 0.86$), followed by energy consultants ($M = 3.87$; $SD = 0.92$), friends with investment experience ($M = 3.76$; $SD = 0.80$), specialized journals ($M = 3.50$; $SD = 0.77$) and the specialized internet forum ($M = 3.24$; $SD = 0.87$).

In 15 of the 17 choice sets all attribute levels were set at random. The other two choice sets were fixed, identical (except for the order of the investment alternatives) and shown twice to all participants.

These *fixed tasks* were meant to measure the respondents answering consistency; high correlations were found between the answers to the fixed tasks ($r=0.76$; $p<0.001$) indicating a high consistency of the participants' responses.

2.2. Measurement of internal factors and further questions

At the beginning of the second part, internal factors were assessed. We considered different factors accounting for different types of dispositions (including egoistic, eco-social and hedonistic orientations) by measuring the decision makers' value orientations. We decided for this measurement as value orientations are by definition quite stable and may influence behavior and decisions across situations (e.g., Refs. [15–18]). Moreover, value orientations have been found to be decision-relevant before (see Table 1).

Value orientations were measured with the short version of the value inventory by Stern et al. [35]. This instrument is based on the work of Schwartz et al. [15–18] and has already been tested in the context of environmentally relevant behaviors (e.g., Refs. [25,35]). The short version contains four scales that are each measured by three items. In comparison to the original version we shortened the response scale from 7 to 5 points ([the value is:] 1 = opposed to my value orientation; 2 = not important; 3 = important; 4 = very important; 5 = extremely important/a guiding principle in my life). The first scale represents *eco-social* (or self-transcendence) *value orientations* covering dispositions towards environment protection, social justice, and keeping peace. The other three scales cover *conservative value orientation* which represents self-discipline and resistance to temptation, family security and honoring elders; *egoistic* (or self-enhancement) *value orientations* which includes seeking for wealth, influence and authority; *hedonistic* (or openness to change) *value orientations* which cover curiosity and seeking for excitement and variety in life.

We calculated the reliabilities for each (three item) value scale: calculations of Cronbach's α revealed acceptable reliability values for eco-social value orientations ($\alpha=0.65$), conservative value orientations ($\alpha=0.65$) and hedonistic value orientations ($\alpha=0.71$). The reliability of the egoistic value scale was rather weak ($\alpha=0.55$) and could not be improved (e.g., by excluding one of the items). Thus, we decided not to consider the egoistic value orientations for further analyses.

For the present study, we expect decision makers with strong eco-social value orientations to be more likely to make solar thermal energy investments—as the use of renewable energies provides eco-social benefits. Prior research supports this assumption (see Table 1; [25]). So far, conservative and hedonistic value orientations have not been studied thoroughly in relation to solar thermal energy investments. However, we assume both strong conservative and hedonistic value orientations to reduce investment likelihood: Decision makers with strong conservative value orientations may be more skeptical towards new technologies, and they may perceive major investments – and spending high amounts of money – as a temptation they need to resist. Strong hedonistic value orientations, on the other hand, may encourage decision makers to spend their money for things other than solar thermal energy investments—namely something offering immediate satisfaction.

In addition to the values scales, the second part of the investigation included questions concerning the perceived recommendation trustworthiness of the five information sources which have been described already.

Finally, socio-demographic data was collected including age, gender, household size, household net income and occupation. The socio-demographic characteristics of our sample were compared to the basic population of German households living in owner-occupied single- and two-family houses (see Table 3). Numbers for the basic population were calculated from data provided by the

Table 3

Socio-demographic sample characteristics in comparison to the German basic population of households living in owner-occupied single and two-family houses.

	Sample (n = 232)	Population ^a
Age		
M	57.4	57.2
SD	8.6	5.2
Gender		
Male	50.0%	49.1%
Female	50.0%	50.9%
Number of household members		
M	2.6	2.5
SD	1.1	0.2
Net household income per month [€]		
M	3651	3851
SD	388	488
Occupation		
Self-Employed (%)	12.5	10.5
Civil servant (%)	3.9	5.2
Employee ^b (%)	31.5	32.5
Worker ^b (%)	4.7	16.6
Retired (%)	34.5	33.5
Unemployed (%)	4.8	1.1
Non-Working ^c (%)	8.2	0.6

^a Approximated values for owner-occupied single and two-family houses calculated from data provided by the German Federal Office of Statistics [5,6,47,48]. The available data did not allow for group comparisons in terms of significance test.

^b Including trainees and apprentices.

^c This category mainly includes housekeeper and students.

German Federal Office of Statistics [5,6,47,48]. The available data did not allow for group comparisons in terms of significance tests. However, face validity indicate our sample to be widely comparable to the basic population in terms of age, gender, household size and income. Considerable differences between our sample and the basic population were found in the occupational status: especially workers were underrepresented in our sample while the number of unemployed and non-working people (e.g., housekeeper) was higher than in the basic population. Thus, the external validity of our findings may be questioned to some degree. A more detailed discussion of potential threats to external validity is provided in Section 3.3.

2.3. Data analyses

Data were analyzed with mixed logit models, an approach particularly suitable to evaluate research designs with repeated choices. Additionally, these models account for different kinds of unobserved factors (see Refs. [39,49] for more information). The models were built with the R package maxLik (see Ref. [50]).

Two different mixed logit models were calculated. In the first model, only main effects of the three internal factors (value scales) and the five external factors (attributes) were calculated in order to verify whether all factors we considered were relevant for households' investment decisions in solar thermal energy systems. In the second model, we added interaction effects between internal and external factors. Only those decision-relevant factors that have been found in the first model were considered here.

3. Results and discussion

3.1. Modell I: verification of internal and external factors' main effects

All three internal factors (eco-social, conservative and hedonistic value orientations) and all five external factors (investment costs, energy cost saving potentials, CO₂ saving potentials,

Table 4
Estimated choice Model I: main effects of internal and external factors.

	Estimate	Std. Error	t
<i>Internal Factors (Value Orientations)</i>			
Conservative Values	−0.60	0.14	−4.37***
Eco-Social Values	0.95	0.15	6.24***
Hedonistic Values	−0.59	0.13	−4.60***
<i>External Factors (Attributes)</i>			
Investment Costs	−0.89	0.04	−24.96***
Energy Cost Saving Potentials	1.00	0.04	26.42***
CO ₂ Saving Potentials	0.35	0.03	10.68***
Trustworthiness of Recommendations	0.29	0.02	14.97***
Guarantee Extend	0.41	0.02	17.03***
Number of respondents	232		
df	3922		
Log Likelihood	3465		
McFadden's Pseudo r ²	0.360		

*** p < 0.001.

guarantee extent and recommendation trustworthiness) entered the model significantly (see Table 4).

Showed the direction of influence we assumed: eco-social value orientations were found to have a positive influence on solar thermal energy investments while conservative value orientations and hedonistic value orientations were both found to have a negative influence. Across the external factors, energy cost saving potentials, CO₂ saving potentials, guarantee extent and trustworthiness of recommendations were found to have a positive influence on investments in solar thermal energy systems. Not surprisingly, influence of investment costs was negative. As all internal and external factors entered the first model significantly all of them were integrated in the second model as well.

3.2. Modell II: main effects and interactions of decision-relevant factors

We calculated the main effects for each internal (eco-social, conservative and hedonistic value orientations) and external factor (investment costs, energy cost saving potentials, CO₂ saving potentials, guarantee extent and recommendation trustworthiness), and interactions between the internal and external factors (see Table 4). Before we turn to the interaction effects, main effects of internal and external factors in the second model will be presented briefly.

Across the internal factors a significant main effect was only found for hedonistic value orientations; as in the first model strong hedonistic value orientations were found to reduce investment likelihood. In this model, no significant main effects were found for conservative and eco-social value orientations. Across the external factors investment costs, energy cost saving potentials and recommendation trustworthiness entered the model significantly. Again high levels of investment costs were found to decrease investment likelihood, while higher levels of energy cost saving potentials and recommendation trustworthiness were found to increase investment likelihood. No significant main effects were found for guarantee extent and CO₂ saving potentials.

Significant interaction effects were found between conservative value orientations and investment costs, conservative value orientations and energy cost saving potentials, and conservative value orientations and trustworthiness of recommendations; no significant interactions were found between conservative value orientations and CO₂ saving potentials or conservative value orientations and guarantee extent. All significant interaction effects are negative indicating that decision makers with strong conservative value orientations are less sensitive towards cost saving potentials and trustworthiness of recommendations but more sensitive

towards investment costs. These interactions seem reasonable as individuals with strong conservative value orientations seek to resist temptations and are self-disciplined. Thus, they may hesitate spending high amounts of money even if there is a chance that making the investment will pay off. Similarly, they may interpret investment recommendations by others as another type of temptation they need to resist—no matter how trustworthy the information source may be.

Eco-social value orientations were found to interact significantly with investment costs, CO₂ saving potentials, and trustworthiness of recommendations but not with energy cost saving potentials and guarantee extent. Here, all significant interactions show positive signs indicating that decision makers with strong eco-social value orientations are more sensitive towards CO₂ saving potentials and trustworthy recommendations while investment costs are comparatively unimportant for their investment decisions. Especially the high relevance of CO₂ saving potentials and trustworthy recommendations is not surprising: individuals with strong eco-social value orientations tend to be concerned with environment protection and social issues; their focus on these issues may also be an explanation why they are less sensitive to financial aspects.

Finally, significant interaction effects were found between hedonistic value orientations and investment costs, energy cost saving potentials, guarantee extent and recommendation trustworthiness. No significant interactions were found between hedonistic value orientations and CO₂ saving potentials. The direction of the significant interactions indicate that decision makers with strong hedonistic value orientations are less sensitive towards investment costs and recommendation trustworthiness but more sensitive towards energy cost saving potentials and guarantee extent. In our view, these findings are not entirely conclusive: On the one hand, individuals with strong hedonistic value orientations seek for excitement and variety. Such value orientation might cause them to be somewhat light-minded when it comes to major investment decisions. Such tendencies may explain why these decision makers are not too sensitive towards investment costs and recommendations. On the other hand, this explanation does not illustrate why they emphasize energy cost saving potentials and guarantee extent. Perhaps, they appreciate investment alternatives promising certain advantages – by not considering the disadvantages thoroughly.

Taken together, we found several interaction effects between internal and external factors influencing solar thermal energy investments significantly. The influence of the main effects decreased as interaction effects were integrated in the decision model. These findings match the assumption that internal and external factors interact in influencing investment decisions in solar thermal energy systems: Some influences may only occur if internal and external factors are combined appropriately.

3.3. Limitations

As for most experimental studies, the external validity of our discrete choice experiment may be threatened to some degree [51]. One important threat of external validity may be based on the fact that we treated the attributes in the decision experiment as independent from each other. However, complete independency is unrealistic in a real-life setting. As we stated before, energy cost saving potentials and CO₂ saving potentials are correlated, for one thing. Additionally, it can be assumed that more expensive solar thermal energy systems often entail higher saving potentials and/or more extensive guarantees. However, we decided for a design with independent (external) factors because our main goal was to compare factor relevance directly—in order to identify the most promising starting points for improving policy measures. For that matter, we found an independent design suitable as pol-

icy makers can target most external factors independently from each other, too. Investment costs can be reduced by subsidies, energy cost saving potentials can be increased if energy costs are raised and guarantees can be extended if regulations are changed accordingly. Naturally, policy measures cannot target CO₂ saving potentials directly. To improve environmental benefits other approaches may be useful—probably by supporting exceedingly eco-friendly manufacturers (e.g., with tax reliefs, certificates). For our decision experiment, however, we found it appropriate to use per cent CO₂ saving potentials to account for environmental benefits. In our view, a (at least in part) comparable attribute scaling might also be easier to understand for the participants. For further experiments of that kind, other designs may be more useful, dependent on the context and purpose of the experiment.

A further threat to the external validity of our study may result from our sample. The comparison of our sample to the basic population indicates that our participants are not entirely representative to households in similar housing situations—especially not in terms of occupation (Table 3). For one thing, the number of non-working (mostly housekeeper) and unemployed persons were comparatively high in our sample. This difference, however, may be due to differences in data collection. In the basic population the occupational status of the “household” was defined by the occupation of the household member with the highest income. On the contrary in the introduction of our study it was pointed out, that the questionnaire should be completed by a person that would participate in an energy-relevant investment decision—which did not need to be the one with the highest income, of course. For instance, the housekeeping partner might be involved in the decision process as well. However, the threat to external validity should not be overestimated. Most energy-relevant investments are collective decisions where different household members are involved [52]. In addition, we found the workers to be underrepresented in our sample—and thus our findings may lack validity for that group. This divergence to the basic population is somewhat hard to explain. Probably, workers are generally less interested in participating in online questionnaires than other groups – but we are not aware of any studies accounting for that assumption. Analyses from further studies may help investigating whether this difference between occupational groups answering online questionnaires is systematic.

Finally, external validity may be threatened because we did not control for the influence of socio-demographic factors in our analyses. In fact, some external factors can be expected to interact with certain socio-demographic factors under real-life circumstances. Especially financial resources (income) may influence the households' sensitivity towards financial incentives. However, we did not control for income as we do not expect it to be an actual, crucial factor for making investments in solar thermal energy systems. Even though purchasing a solar thermal energy system is a major investment the total amount of money to be spent rarely exceeds € 10,000—which is low compared to other kinds of energy-relevant investments (e.g., major insulation measures; or purchasing a new car [4,22]). This amount can hardly be expected to be a highly important barrier—especially as most households living in single- and two-family houses belong to a high income class (see Table 3; [5,6,47]) and as financing models at very low interest rates for such-like investments are available [53]. Previous studies investigating the decision-relevance of income on energy-relevant investment decisions support our approach (see Ref. [34] for an overview). Especially for investments in solar thermal energy systems findings concerning the role of income are inconclusive so far: while some investigations indicate a high income having a positive effect on solar thermal energy investments, others show it to be irrelevant or to even have a negative effect. For other energy-relevant

investment decisions such as certain insulation measures, on the other hand, income was found to be more crucial.

4. Conclusions and policy implications

The goal of our analyses was to identify starting points for measures motivating home owners in Western Countries to make investments in renewable energy technology in their households. Against the background of recent empirical studies (see Ref. [34] for an overview) and based on theories about the interplay of internal and external determinants of environmentally relevant decisions (e.g., Refs. [14,21]), we developed the assumption that internal factors (i.e. decision makers' dispositions) may interact with the “classical” external factors (i.e. expected financial and environmental consequences, expected operation security, and expected social support and investment recommendations by others). Contrary to former investigations (which mostly focused on eco-social dispositions as internal factors only), we considered several different types of dispositions that can be expected to be decision-relevant, including conservative, eco-social and hedonistic (value) orientations. Our initial analyses confirmed the decision relevance of several internal factors (in terms of the value orientations) as well as of the external factors we recorded (investment costs, energy cost saving potentials, CO₂ saving potentials, trustworthiness of recommendations, guarantee extent). However the pattern became clearer when we allowed for interactions between the internal and external factors.

The more differentiated analyses revealed that the strength of the conservative value orientation had no overall direct impact at all, only in interaction with external factors, in particular investment costs and trustworthiness of recommendations. The extent of the eco-social value orientations also had no direct effect but showed strong interaction effects with expected CO₂ saving potentials and trustworthiness of recommendations. Hedonistic value orientations had an overall main effect and interacted with nearly all external factors—except CO₂ saving potentials. Not surprisingly, environmental consequences seem not relevant for individuals scoring high on this value scale.

This pattern shows that consequences of investment decisions in renewable energy technology are of different relevance for individuals with differing value orientations. While investment costs, cost saving potentials and trustworthiness of recommendations seem of high relevance to *all* decision makers, environmental consequences and operational security may only be relevant for subgroups with a strong eco-social value orientation or strong hedonistic value orientations. Trustworthiness of recommendation was found to be an important positive external factor (craft businesses were rated to be most trustworthy followed by energy consultants, friends, specialized journals and internet forums), and should be given more consideration.

Thus overall, the results broaden the current view on household decisions and reveal starting points for the adjustment of measures – or packages of measures – to promote renewable energy systems for homeowners: first of all – and not surprisingly – the study supports the importance of reducing investment costs, which can be done with subsidies or reduced loans like it is already common in several (mostly Western) countries [53]. Moreover, the findings point to the relevance of expected energy cost savings. Hence, it may be a problem that energy prices for the main domestic heat sources (fossil fuels) are stable or decreasing in most Western Countries [54]. As a consequence, energy cost saving potentials for heat consumption are limited—especially as compared to domestic electricity consumption where prices are widely increasing. It becomes evident that especially low energy prices for fossil fuels have to be countered by respective measures, e.g., environmental taxes.

Earnings from such measures could be used to increase incentives for renewable heat energies. Nevertheless, a solely incentive based strategy seems to waste a relevant part of the potential for fostering investment decisions.

For one thing, the study confirms the important role of trust and trustworthiness of different sources for investment recommendations. Overall, craft businesses were rated to be the most trustworthy – and thus the most influential – information source within the decision process. This finding claims to further develop and test measures to enhance trustworthiness of different actors in the field: At this, certification procedures may be helpful in particular. Craft businesses could be certified if staff members received professional trainings concerning renewable energies. Besides, closer cooperation between craft businesses and professional energy advisers could be initiated. Craft businesses that are not sufficiently trained for energy consulting could recommend professional energy advisers if they are approached by households considering energy-relevant investments; incentives for such recommendations could be established as well (e.g., conciliatory premiums for the recommending craft businesses). Standardizing the certification of energy advisers (e.g., across the whole European Union) could be an additional opportunity to increase homeowners' trust in a relevant source of information.

For another thing, the study points to the necessity to differentiate between target groups and calls for the development of tailored packages of measures and their tailored marketing. For certain target groups it might be advisable not to focus on financial incentives but on other measures. Our analyses show, for instance, that home owners with strong hedonistic value orientations are more interested in guarantees but do not care too much for investment costs. For such target groups it may be worthwhile offering investment alternatives with broad guarantees where investment costs can probably be higher.

Additionally, different communication measures could be used for different target groups. Homeowners with strong conservative and hedonistic value orientations should be approached by mass media (e.g., specialized journals) as they are not sensitive towards (personal) recommendations. Homeowners with strong eco-social value orientations, on the other hand, may better be addressed by direct marketing. Using different communication channels the marketing could also highlight different (target group-specific) concerns and opportunities—e.g., reduction of CO₂ emissions for home owner with strong eco-social value orientations, or possibilities to get extended guarantees for home owners with strong hedonistic value orientations. Recent studies and discussions (e.g., Refs. [55,56]) have focused on positive or negative spillover effects between different forms of pro-environmental consumer behavior and policy acceptance. In the light of these discussions it seems advisable to always communicate both, the positive financial (saving energy costs) and environmental (reducing CO₂ emissions) consequences of an investment decision for renewable energy. In particular with respect to the long term strategy of several European countries for an extensive reduction of CO₂ emissions in the household sector the findings point to long term measures (i.e. broad educational measures) to enhance in general a stronger eco-social value orientation.

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References

- [1] Intergovernmental Panel on Climate Change (IPCC), Mitigation of Climate Change. Summary for Policymakers, 2014, Retrieved from <https://www.ipcc.ch/report/ar5/wg3/> (10.04.14).
- [2] European Commission, EU Energy in Figures—Statistical Pocketbook 2014, 2014, Retrieved from http://ec.europa.eu/energy/sites/ener/files/documents/2014_pocketbook.pdf (17.03.15).
- [3] International Energy Agency, Worldwide Energy Trends, 2008, Retrieved from http://www.iea.org/publications/freepublications/publication/Indicators_2008.pdf (12.11.15).
- [4] Deutsche Energie-Agentur, Modernisierungsratgeber Energie—Kosten sparen, Wohnwert steigern, Umwelt schonen [Modernization Guide Energy—Saving Costs, Increasing the House Value, Protecting the Environment], Schöneweide, 2010.
- [5] Statistisches Bundesamt [German Federal Office of Statistics], Wirtschaftsrechnungen—Einkommens- und Verbrauchsstichprobe Wohnverhältnisse privater Haushalte [Economic Accounts], 2015, Retrieved from https://www.destatis.de/DE/Publikationen/Thematisch/EinkommenKonsumLebensbedingungen/EinkommenVerbrauch/EVS_EinnahmenAusgabenprivaterHaushalte2152604139004.pdf?__blob=publicationFile (25.11.15).
- [6] Statistisches Bundesamt [German Federal Office of Statistics], Wirtschaftsrechnungen—Einkommens- und Verbrauchsstichprobe Wohnverhältnisse privater Haushalte [Economic Accounts], 2013, Retrieved from https://www.destatis.de/DE/Publikationen/Thematisch/EinkommenKonsumLebensbedingungen/EinkommenVerbrauch/EVS_HausGrundbesitzWohnverhaeltnisHaushalte2152591139004.pdf?__blob=publicationFile (25.11.05).
- [7] Verbraucherzentrale Nordrhein-Westfalen e.V. [Consumer Advice Center North Rhine-Westphalia], Solarwärme für Warmwasser und Heizung [Solar Heat for Hot Water and Heating], 2015, Retrieved from <http://www.vz-nrw.de/solarthermie> (17.03.15).
- [8] R. Corradini, Regional differenzierte Solarthermie-Potenziale für Gebäude mit einer Wohneinheit [Regional potentials for solar thermal energy in buildings with one residential unit], 2013.
- [9] Bundesministerium für Verkehr Bau und Stadtentwicklung, CO₂ Gebäudereport 2007 [CO₂ Building Report], vier C print+mediafabrik, Berlin, 2007.
- [10] A.B. Jaffe, R.N. Stavins, The energy-efficiency gap. What does it mean? *Energy Policy* 22 (10) (1994) 804–810.
- [11] Bundesministerium für Wirtschaft und Energie [Federal Ministry for Economic Affairs and Energy], Erneuerbare Energien im Jahr 2014 [Renewable Energies in the year 2014], 2015, Retrieved from <https://www.bmwi.de/BMWi/Redaktion/PDF/Publikationen/erneuerbare-energien-im-jahr-2014,property=pdf,bereich=bmwi2012,sprache=de,rwb=true.pdf> (17.03.15).
- [12] Eurostat, Distribution of population by tenure status, type of household and income group, 2015, Retrieved from [http://www.census.gov/housing/hvs/files/annual14/ann14t_14.xls](http://appsso.eurostat.ec.europa.eu/nui/show.do?query=BOOKMARK_DS-057102_QID_684D63DF.UID-3F171EB0&layout=TIME,C,X,0;GEO,L,Y,0;INCGRP,L,Z,0;HHTYP,L,Z,1;TENURE,L,Z,2;INDICATORS,C,Z,3;&zSelection=DS-057102INDICATORS,OBS-FLAG;DS-057102INCGRP,TOTAL;DS-057102TENURE,OWN;DS-057102HHTYP,TOTAL;&rankName1=HHTYP.1.2.-1.2&rankName2=INDICATORS.1.2.-1.2&rankName3=INCGRP.1.2.-1.2&rankName4=TENURE.1.2.-1.2&rankName5=TIME.1.0.0.0&rankName6=GEO.1.2.0.1&sortR=ASC.0&sortC=ASC.-1.FIRST&rStp=&cStp=&rDCh=&cDCh=&rDM=true&CDM=true&footnes=false&empty=false&wai=false&time.mode=NONE&time.most_recent=false&lang=EN&cfo=%23%23%23%2C%23%23%23.%23%23%23 (12.11.15).
[13] United States Census, Homeownership Rates by Area, 2015, Retrieved from <a href=) (12.11.15).
- [14] G.A. Guagnano, P.C. Stern, T. Dietz, Influences on attitude-behavior relationships: a natural experiment with curbside recycling, *Environ. Behav.* 27 (5) (1995) 699–718.
- [15] S.H. Schwartz, W. Bilsky, Toward a theory of the universal content and structure of values: extensions and cross-cultural replications, *J. Personal. Soc. Psychol.* 58 (5) (1990) 878.
- [16] S.H. Schwartz, Universals in the content and structure of values: theoretical advances and empirical tests in 20 countries, *Adv. Exp. Soc. Psychol.* 25 (1) (1992) 1–65.
- [17] W. Bilsky, S.H. Schwartz, Values and personality, *Eur. J. Personal.* 8 (3) (1994) 163–181.
- [18] S.H. Schwartz, Are there universal aspects in the structure and contents of human values? *J. Soc. Issues* 50 (4) (1994) 19–45.
- [19] P.C. Stern, Toward a coherent theory of environmentally significant behavior, *J. Soc. Issues* 56 (3) (2000) 407–424.
- [20] A. Kollmuss, J. Agyeman, Mind the gap: why do people act environmentally and what are the barriers to pro-environmental behavior? *Environ. Educ. Res.* 8 (3) (2002) 239–260.

- [21] A. Diekmann, P. Preisendörfer, *Umweltsoziologie [Environmental Sociology]*, Rowohlt, Reinbek bei Hamburg, 2001.
- [22] Bausparkasse Schwäbisch Hall [Building Association Schwäbisch Hall], *Renovierungskosten berechnen [Calculating Renovation Costs]*, 2015, Retrieved from <https://www.schwaebisch-hall.de/ham/bauen-modernisieren/neubau-umbau/artikel/h1103.Kosten-fuer-Modernisierung.php> (05.02.15).
- [23] K. Wortmann, *Psychologische Determinanten des Energiesparens [Psychological Determinants of Saving Energy]*, vol. 21, Beltz Psychologie Verlags Union, Weinheim, 1994.
- [24] M. Herzog, *Psychologische und ökonomische Aspekte bei der Förderung von regenerativen Energieträgern [Psychological and economic aspects of the promotion of renewable energy sources]*, Lang, 1997.
- [25] I. Kastner, E. Matthies, M. Willenberg, *Chancen zur Förderung nachhaltigkeitsrelevanter Investitionsentscheidungen durch psychologisch basiertes Framing—eine Pilotstudie. [Prospects of increasing sustainability-relevant investment decisions through psychologically based framing—a pilot study]*, *Umweltpsychologie* 15 (1) (2011) 30–51.
- [26] H. Welsch, J. Kühling, *Determinants of pro-environmental consumption: the role of reference groups and routine behavior*, *Ecol. Econ.* 69 (1) (2009) 166–176.
- [27] D.G. Labay, T.C. Kinnear, *Exploring the consumer decision process in the adoption of solar energy systems*, *J. Consum. Res.* (1981) 271–278.
- [28] M. Achtnicht, R. Madlener, *Factors influencing German house owners' preferences on energy retrofits*, *Energy Policy* 68 (2014) 254–263.
- [29] G. Hübner, M. Müller, *Erneuerbare Energien und Ökostrom—zielgruppenspezifische Kommunikationsstrategien. Modul I—Analyse der Konsumentenentscheidungen für Erneuerbare Energien und Ökostrom [Renewable Energies and Green Power—target group specific communication strategies. Module I—Analysis of Consumer Decisions regarding Renewable Energies and Green Power]*, 2012, Retrieved from <http://sozpsy-forschung.psych.uni-halle.de/workshopEEOSZK/projektbericht-EE-kommunizieren-modul-i-langfassung-2012.pdf> (15.05.13).
- [30] M. Sadler, *Home Energy Preferences & Policy: Applying Stated Choicemodeling to a Hybrid Energy Economy Model*, Simon Fraser University, 2003, Retrieved from (21.10.15).
- [31] K. Mahapatra, L. Gustavsson, *An adopter-centric approach to analyze the diffusion patterns of innovative residential heating systems in Sweden*, *Energy Policy* 36 (2) (2008) 577–590.
- [32] C.C. Michelsen, R. Madlener, *Homeowners' preferences for adopting residential heating systems: a discrete choice analysis for Germany*, *Energy Econ.* 34 (5) (2011) 1271–1283.
- [33] L. Murphy, *The influence of energy audits on the energy efficiency investments of private owner-occupied households in the Netherlands*, *Energy Policy* 65 (2014) 398–407.
- [34] I. Kastner, P.C. Stern, *Examining the decision-making processes behind household energy investments: a review*, *Energy Res. Soc. Sci.* 10 (2015) 72–89.
- [35] P.C. Stern, T. Dietz, G.A. Guagnano, *A brief inventory of values*, *Educ. Psychol. Meas.* 58 (6) (1998) 984–1001.
- [36] G.T. Gardner, P.C. Stern, *Environmental Problems and Human Behavior*, Pearson, Boston, MA, 2002.
- [37] K. Backhaus, B. Erichson, R. Weiber, *Fortgeschrittene multivariate Analysemethoden, in: Eine Anwendungsorientierte Einführung [Advanced Multivariate Analysis Methods. An Application-oriented Introduction]*, Springer, Heidelberg, Berlin, 2011.
- [38] W.H. Greene, *Econometric Analysis*, Granite Hill Publishers, 2008.
- [39] D. Hoyos, *The state of the art of environmental valuation with discrete choice experiments*, *Ecol. Econ.* 69 (8) (2010) 1595–1603.
- [40] T.D. Cook, D.T. Campbell, A. Day, *Quasi-Experimentation: Design & Analysis Issues for Field Settings*, vol. 351, Houghton Mifflin, Boston, 1979.
- [41] H.A. Simon, *A behavioral model of rational choice*, *Q. J. Econ.* 69 (1) (1955) 99–118.
- [42] H.A. Simon, *Rationality as process and as product of thought*, *Am. Econ. Rev.* 68 (2) (1978) 1–16.
- [43] Bauherren-Schutzbund [Builder Protection Association], *Angebotscheck für Photovoltaikanlagen [Checking offers for photovoltaic systems]*, 2013, Retrieved from http://www.bsb-ev.de/fileadmin/user_upload/ratgeber_aktuell/Ratgeber_aktuell.31.Photovoltaik.pdf (05.02.15).
- [44] R.D. Miller, J.M. Ford, *Shared Savings in the Residential Market: A Public/Private Partnership for Energy Conservation*, Energy Task Force, Urban Consortium for Technology Initiatives, Baltimore, MD, 1985.
- [45] P.C. Stern, E. Aronson, J.M. Darley, D.H. Hill, E. Hirst, W. Kempton, T.J. Wilbanks, *The effectiveness of incentives for residential energy conservation*, *Eval. Rev.* 10 (2) (1986) 147–176.
- [46] Z.L. Tormala, P. Brinol, R.E. Petty, *When credibility attacks: the reverse impact of source credibility on persuasion*, *J. Exp. Soc. Psychol.* 42 (5) (2006) 684–691.
- [47] Statistisches Bundesamt [German Federal Statistical Office], *Statistisches Jahrbuch [statistical yearbook]*, 2014, Retrieved from http://www.die-stromsparinitiative.de/fileadmin/bilder/Stromspiegel/Brosch%C3%BCre/Stromspiegel2014web_final.pdf (31.08.15).
- [48] Statistisches Bundesamt [German Federal Office of Statistics], *Bevölkerung und Erwerbstätigkeit [Population and Occupation]*, 2015, Retrieved from https://www.destatis.de/DE/Publikationen/Thematisch/Bevoelkerung/Bevoelkerungsstand/VorlBevoelkerungsfortschreibung5124103149004.pdf?__blob=publicationFile (25.11.15).
- [49] K.E. Train, *Discrete Choice Methods with Simulation*, Cambridge University Press, Cambridge, 2003.
- [50] A. Henningsen, O. Toomet, *maxLik: a package for maximum likelihood estimation in R*, *Comput. Stat.* 26 (3) (2011) 443–458.
- [51] J. Bortz, N. Döring, *Forschungsmethoden und Evaluation: für Human-und Sozialwissenschaftler [Research Methods and Evaluation: for Humanities and Social Sciences]*, Springer, Heidelberg, 2006.
- [52] B.M. Sopha, C.A. Klöckner, E.G. Hertwich, *Adopters and non-adopters of wood pellet heating in Norwegian households*, *Biomass Bioenergy* 35 (1) (2011) 652–662.
- [53] International Energy Agency, *Addressing Climate Change. Policies and Measures Databases*, 2014, Retrieved from <http://www.iea.org/policiesandmeasures/climatechange/> (30.10.14).
- [54] International Energy Agency, *Energy Prices and Taxes*, 2015, Retrieved from <http://www.oecd-ilibrary.org/docserver/download/6215031e.pdf?expires=1449145110&id=id&accname=ocid54022140&checksum=58f6dbfe56b587517e832592b575f677> (03.12.15).
- [55] J. Thøgersen, T. Crompton, *Simple and painless? The limitations of spillover in environmental campaigning*, *J. Consum. Policy* 32 (2) (2009) 141–163.
- [56] J. Thøgersen, C. Noblet, *Does green consumerism increase the acceptance of wind power? Energy Policy* 51 (2012) 854–862.

Update

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Corrigendum

Corrigendum to “Investments in renewable energies by German households: A matter of economics, social influences and ecological concern?” [Energy Res. Soc. Sci. 17 (2016) 1–9]



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In Section 3.2, [Table 5](#) referring to the *Estimated choice Model II: Main effects and interactions of decision-relevant variables* is missing. Mistakenly, another reference to Table 4 (describing model I) is given in this section.

The missing table is presented below:

Table 5
Estimated Choice Model II: Main effects and interactions of decision-relevant variables.

	Estimate	Std. Error	t
Main effects			
<i>Internal Factors (Value Orientations)</i>			
Conservative Values	0.08	0.34	0.24
Eco-Social Values	−0.34	0.34	−1.01
Hedonistic Values	−0.91	0.30	−3.04**
<i>External Factors (Attributes)</i>			
Investment Costs	−1.56	0.30	−5.27**
Energy Cost Saving Potentials	1.09	0.30	3.68**
CO ₂ Saving Potentials	−0.33	0.27	−1.23
Trustworthiness of Recommendations	0.48	0.17	2.87**
Guarantee extend	−0.16	0.20	−0.80
Interactions			
Conservative Values × Investment Costs	−0.16	0.07	−2.44*
Conservative Values × Energy Cost Saving Potentials	−0.14	0.07	−1.97*
Conservative Values × CO ₂ Saving Potentials	−0.08	0.07	−1.16
Conservative Values × Guarantee Extent	−0.02	0.05	−0.33
Conservative Values × Trustworthiness of Recommendations	−0.14	0.04	−3.69**
Eco-Social Values × Investment Costs	0.18	0.07	2.48*
Eco-Social Values × Energy Cost Saving Potentials	−0.08	0.07	−1.09
Eco-Social Values × CO ₂ Saving Potentials	0.21	0.07	2.87**
Eco-Social Values × Guarantee Extent	0.02	0.05	0.44
Eco-Social Values × Trustworthiness of Recommendations	0.18	0.04	4.28**
Hedonistic Values × Investment Costs	0.19	0.06	3.33**
Hedonistic Values × Energy Cost Saving Potentials	0.29	0.06	4.63**
Hedonistic Values × CO ₂ Saving Potentials	0.04	0.06	0.78
Hedonistic Values × Guarantee Extent	0.19	0.04	4.48**
Hedonistic Values × Trustworthiness of Recommendations	−0.11	0.03	−3.10**
Number of respondents	232		
df	3907		
Log Likelihood	−3429		
McFadden's Pseudo r ²	0.367		

* p < 0.05.

** p < 0.01.

*** p < 0.001.

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